

Highlights

Six-degree-of-freedom cascaded fractional-order control of an over-actuated ROV: margin-constrained tuning and a bias-controlled comparison of metaheuristics

Hakan Ersoy, [\[TODO: co-authors, to be confirmed\]](#)

- Cascaded FOPID-(1+TFOID) control of all six DOF of an over-actuated BlueROV2 Heavy
- Phase margins are imposed over a range of thruster lag and latency during tuning
- Tuning on nominal cases trades robustness for nominal cost in every cascade
- With robust tuning, the fractional cascade beats the P-PID cascade by 35% in storms
- Equal budgets and a mirrored search box expose the origin bias of SOO and GJO

Six-degree-of-freedom cascaded fractional-order control of an over-actuated ROV: margin-constrained tuning and a bias-controlled comparison of metaheuristics

Hakan Ersoy^{a,*}, [TODO: co-authors, to be confirmed]^a

^a*Department of Electronics and Communications Engineering, Yıldız Technical University, İstanbul, 34220, Türkiye*

Abstract

Fractional-order controllers for remotely operated vehicles are mostly designed for one axis, with idealized thrusters and no latency. We recast the cascaded FOPID-(1+TFOID) controller, proposed for load-frequency control, as a position–velocity cascade for all six degrees of freedom of an over-actuated BlueROV2 Heavy with eight T200 thrusters, whose thrust, dead band, saturation and electrical power are fitted to the manufacturer’s measurements. Tuning minimizes a multi-scenario six-DOF ITAE and energy cost subject to phase margins on every linearized loop over a range of thruster lag (0.05–0.2 s) and latency (10–60 ms). Six metaheuristics, including the recent stellar oscillation optimizer (SOO), are compared at an equal evaluation budget and in a mirrored search box: particle swarm optimization gives the lowest median cost, unchanged in the mirrored box, while SOO and golden jackal optimization are attracted to the coordinate origin. Over 120 tuning runs on nominal cases, a lower training cost goes with more divergences in a 100-draw Monte Carlo study of plant and actuator parameters,

*Corresponding author.

for every cascade. With a robust training set that adds a severe-noise case and a weak-actuator case, none of the 40 cascade tunings diverges, and the proposed cascade is the best structure: its median training cost is 36% below that of a tuned integer P-PID cascade ($p = 0.026$), with 35% less error in a multi-set-point storm test and three times less under severe sensor noise. Tuning without latency and margin constraints gives margins down to 3.8° and a storm-test cost 1.8–5.6 times higher at 30 ms of latency.

Keywords: remotely operated vehicle, fractional-order control, cascade control, FOPID, tilt-integral-derivative, metaheuristic tuning, phase margin, BlueROV2

1. Introduction

Remotely operated vehicles move in six coupled degrees of freedom with nonlinear damping, added-mass and restoring effects, and are disturbed by waves and currents [1]. Their thrusters add a further layer: a dead band, asymmetric and voltage-dependent saturation, and motor and propeller dynamics that limit the achievable bandwidth [2, 3]. Small open platforms such as the BlueROV2 have made experiments affordable, and open benchmark models of its heavy configuration exist [4]; with eight thrusters for six degrees of freedom, that configuration is also over-actuated.

Fractional-order $PI^\lambda D^\mu$ controllers [5] have been applied to depth, yaw and motion control of underwater vehicles [6, 7, 8, 9], typically in simulation, axis by axis and with simplified actuators. In our earlier study [10] a $PI^\lambda D^\mu$ surge-speed controller was tuned by PSO [11] and DE [12] over a multi-scenario cost on a fourth-order linear thruster model identified from public

T200 data, and realized on an FPGA. That study was limited to one axis, a linear thruster model and nominal tuning, and named multi-DOF control and the thruster nonlinearity as the next steps.

In load-frequency control, cascades of fractional-order stages, such as PI-(1+FOPID) [13], FOPID-FOPI [14], FOPI-FOPD [15] and, most recently, FOPID-(1+TFOID) [16], have been reported to outperform single-stage controllers. In FOPID-(1+TFOID), the second stage acts on the output of the first minus the local frequency deviation: a local feedback inside the cascade, which for a vehicle is the position-velocity cascade of marine control practice. In [16] the controller is tuned on one nominal scenario without stability margins, and the authors note that their stability evidence is numerical.

The tuners are often the newest metaheuristics, here the stellar oscillation optimizer (SOO) [17]. The value of metaphor-based methods has been questioned [18, 19], and several are attracted to the centre or origin of the search box, which flatters them on benchmarks whose optimum lies there [20]. Comparisons at equal iteration counts also give methods that evaluate twice per iteration, such as SOO, twice the budget, and the SOO paper and its released code disagree in several places [21]. Robustness constraints are classical in PID tuning [22] but rarely part of metaheuristic tuning of fractional-order controllers.

The contributions of this paper are:

1. The FOPID-(1+TFOID) cascade is recast as a position-velocity cascade (the second stage acts on $u_1 - \nu$, the mechanical counterpart of $u_1 - \Delta f$) and applied to all six DOF of an over-actuated BlueROV2 Heavy, with six benchmark structures realized in the same cascade

form.

2. Tuning is subject to phase margins on every linearized DOF loop, imposed over a range of thruster lag and latency because the T200 dynamics are not known well. Without them, ITAE tuning returns loops with margins below 10° that one or two extra samples of latency destabilize (Sect. 6.6).
3. We show that tuning on nominal cases trades robustness for nominal cost: over 120 runs, a lower nominal training cost goes with more divergences in a Monte Carlo study, for every cascade. Adding a severe-noise case and a weak-actuator case to the training set removes the divergences of all structures; under this robust tuning the proposed cascade is the best structure, significantly better than the integer P-PID cascade.
4. The thruster model (thrust, dead band, asymmetry, saturation and electrical power at 10–20 V) is fitted to the manufacturer’s measurements [23], so the effort term is energy in joules and battery voltage is a physically meaningful uncertainty.
5. Six tuners, including the newest (SOO), are compared at an equal evaluation budget, in the original and in a mirrored search box, and on shifted benchmark functions; this separates search ability from attraction to the coordinate origin.

Section 2 describes the vehicle and thruster model, Sect. 3 the controllers, Sect. 4 the tuning problem and Sect. 5 the tuners and the comparison protocol. Section 6 reports the results, followed by a discussion of the limitations and the conclusions.

Table 1: BlueROV2 Heavy parameters [4].

mass m	13.5 kg	volume ∇	0.0135 m ³ (neutral)
I_x, I_y, I_z	0.26, 0.23, 0.37 kg m ²	$\mathbf{r}_b$	[0, 0, -0.01] m
added mass	$X_{\dot{u}}$ 6.357, $Y_{\dot{v}}$ 7.121, $Z_{\dot{w}}$ 18.69 kg; $K_{\dot{p}}$ 0.1858, $M_{\dot{q}}$ 0.1348, $N_{\dot{r}}$ 0.2215 kg m ²		
linear damping	X_u 13.7, Y_v 0, Z_w 33 N s/m; K_p 0, M_q 0.8, N_r 0 N m s/rad		
quadratic damping	$X_{u u }$ 141, $Y_{v v }$ 217, $Z_{w w }$ 190 N s ² /m ² ; $K_{p p }$ 1.192, $M_{q q }$ 0.47, $N_{r r }$ 1.5 N m s ² /rad ²		

2. Vehicle and actuator model

2.1. Rigid-body and hydrodynamic model

Let $\boldsymbol{\eta} = [x, y, z, \phi, \theta, \psi]^\top$ be the position and attitude in the north-east-down frame and $\boldsymbol{\nu} = [u, v, w, p, q, r]^\top$ the velocities in the body frame, whose origin is the centre of gravity. The vehicle obeys [1]

$$\begin{aligned} \mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{g}(\boldsymbol{\eta}) &= \mathbf{T}\mathbf{f} + \boldsymbol{\tau}_d, \\ \dot{\boldsymbol{\eta}} &= \mathbf{J}(\boldsymbol{\eta})\boldsymbol{\nu}, \end{aligned} \quad (1)$$

where $\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$ is diagonal, $\boldsymbol{\nu}_r = \boldsymbol{\nu} - [\mathbf{R}^\top \mathbf{v}_c; \mathbf{0}]$ is the velocity relative to an irrotational current $\mathbf{v}_c$ (the term $\mathbf{M}_A \dot{\mathbf{v}}_c$ is neglected), $\mathbf{D}(\boldsymbol{\nu}_r) = \mathbf{D}_l + \mathbf{D}_q \text{diag}(|\boldsymbol{\nu}_r|)$, and $\mathbf{g}$ contains the restoring forces of the weight and buoyancy. The parameters of the heavy configuration are those of the open benchmark of [4] (Table 1).

2.2. Thrusters and allocation

Four horizontal thrusters at 45° and four vertical thrusters give the 6×8 allocation matrix $\mathbf{T}$ of [4]. A demanded generalized force $\boldsymbol{\tau}$ is distributed by $\mathbf{f}_d = \mathbf{T}^+ \boldsymbol{\tau}$ and scaled down as a whole when a thruster limit is reached, so the direction of $\boldsymbol{\tau}$ is kept. Each command is mapped to a normalized PWM

value $u_i \in [-1, 1]$ by the inverse of the nominal 16 V curve, with dead-band compensation. The thrust produced is

$$F(u) = \begin{cases} \sum_{k=1}^3 a_k (u - d_+)^k, & u > d_+, \\ 0, & -d_- \leq u \leq d_+, \\ -\sum_{k=1}^3 b_k (-u - d_-)^k, & u < -d_-, \end{cases} \quad (2)$$

followed by a first-order lag τ_m . No reliable dynamic model of the T200 is available: the manufacturer estimates 25–40 ms of ESC latency for a running motor, and about 110 ms has been reported from standstill. We therefore take $\tau_m = 0.1$ s and a latency of 30 ms as nominal values and design for $\tau_m \in [0.05, 0.2]$ s and latencies of 10–60 ms (Sect. 4). The coefficients were fitted by least squares to the manufacturer’s measurements at 10–20 V [23]; at 16 V the fit error is 0.24 N RMS (1.3 N maximum) over the range -39.9 N to 51.4 N. The electrical power of the actual thrust, $P(F) = \sum_k q_k |F|^k$, is fitted to the same data (3.1 W RMS of 389 W). The command reaches the thrusters three control periods ($3\Delta t = 30$ ms) after the measurement it was computed from.

2.3. Disturbances and navigation errors

Waves act as a sum of five sinusoids per DOF with random frequencies in 0.4 rad/s to 1.6 rad/s (amplitudes 20, 20, 15 N and 1 N m), and a current of 0.25 m/s ($\pm 20\%$ slow variation) acts through $\boldsymbol{\nu}_r$. The controller sees the output of a navigation filter, modelled as first-order Gauss–Markov errors with 2 Hz bandwidth (one sigma: position 1 cm, depth 5 mm, attitude 0.17–0.29°, velocity 1 cm/s, rates 0.29°/s).

3. Controller

3.1. Position-velocity form of the FOPID-(1+TFOID)

For each DOF i , the first stage acts on the position error $e_{1,i}$ (expressed in the body frame for the translational DOFs) and returns a velocity reference,

$$u_{1,i}(s) = \left(K_p + \frac{K_{i1}}{s^{\lambda_1}} + K_{d1}s^{\mu_1} \right) e_{1,i}(s). \quad (3)$$

The second stage acts on $e_{2,i} = u_{1,i} - \nu_i$ and returns an acceleration demand,

$$a_i(s) = \left(1 + \frac{K_t}{s^{1/n}} + \frac{K_{i2}}{s^{\lambda_2}} + K_{d2}s^{\mu_2} \right) e_{2,i}(s), \quad \tau_i = M_{ii} a_i. \quad (4)$$

In [16] the second stage acts on $u_1 - \Delta f$; here the local feedback is the measured velocity, which makes the cascade a position-velocity cascade in the usual sense of marine control. Scaling by the nominal inertia M_{ii} makes the gains dimensionally consistent across DOFs. One parameter set is shared by (x, y, z) and one by (ϕ, θ, ψ) : 22 decision variables.

3.2. Benchmark structures

Table 2 lists the six benchmarks. FOPID is the single-loop $\text{PI}^\lambda \text{D}^\mu$ of [10] extended to six DOF, P-PID the integer cascade of common ROV autopilots, and the three fractional cascades are those compared in [16]. All cascades use the position-velocity form of Eqs. (3)–(4), so the comparison isolates the stage transfer functions.

3.3. Realization

Each s^α is approximated by the Oustaloup filter [24] of order $N = 5$ on 10^{-3} rad/s to 10^2 rad/s ; for $\alpha < -1$ an exact integrator is split off. Every first-order section is discretized by the Tustin rule at $\Delta t = 10 \text{ ms}$. Integer

Table 2: Controller structures. G_1 acts on the position error; G_2 on $u_1 - \nu$ (cascades only).

Decision variables: two groups (translation, rotation).

structure	$G_1(s)$	$G_2(s)$	variables
FOPID-(1+TFOD) (proposed)	$K_p + K_{i1}s^{-\lambda_1} + K_{d1}s^{\mu_1}$	$1 + K_ts^{-1/n} + K_{i2}s^{-\lambda_2} + K_{d2}s^{\mu_2}$	22
FOPID [10]	$K_p + K_is^{-\lambda} + K_ds^\mu$	–	10
PID	$K_p + K_i/s + K_ds/(s/\omega_h + 1)$	–	6
P-PID	K_{p1}	$K_{p2} + K_{i2}/s + K_{d2}s/(s/\omega_h + 1)$	8
PI-(1+FOPID)	$K_{p1} + K_{i1}/s$	$1 + K_{p2} + K_{i2}s^{-\lambda} + K_{d2}s^\mu$	14
FOPID-FOPi	$K_{p1} + K_{i1}s^{-\lambda_1} + K_{d1}s^{\mu_1}$	$K_{p2} + K_{i2}s^{-\lambda_2}$	16
FOPi-FOPD	$K_{p1} + K_is^{-\lambda}$	$K_{p2} + K_ds^\mu$	12

derivatives use the same high-frequency limit, $K_ds/(s/\omega_h + 1)$ with $\omega_h = 100$ rad/s. The reference passes a critically damped second-order model with $\omega = 1$ rad/s, identical for all structures.

4. Tuning problem

4.1. Cost

The *nominal* training set has two cases, whose disturbance and noise realizations differ from those of the tests: a simultaneous step in x , y , z and ψ (20 s), and station keeping in waves and current with steps in ψ and z (30 s). The *robust* training set adds a step under severe white sensor and process noise (the noise model of test T2, another realization, 15 s) and a corner case in waves with a weak actuator and a heavier vehicle (12 V battery, $\tau_m = 0.2$ s, 60 ms, added mass $\times 1.5$, 20 s). The cost is

$$J = \sum_s \left[\sum_{i=1}^6 w_i \int_0^{T_s} t |e_i(t)| dt + \rho E_s \right] + K_{pm} \sum_a \sum_{i=1}^6 \frac{\max(0, \text{PM}_a^* - \text{PM}_{a,i})}{\text{PM}_a^*}, \quad (5)$$

with $w = [1, 1, 1, 2, 2, 2]$ (one metre of error weighs as much as 0.5 rad), the electrical energy E_s in joules, $\rho = 2 \times 10^{-4} \text{ J}^{-1}$ and $K_{pm} = 200$. The margins are required at three actuator points a : the nominal thruster ($\tau_m = 0.1 \text{ s}$, 30 ms; $\text{PM}^* = 45^\circ$) and the slow (0.2 s, 60 ms) and fast (0.05 s, 10 ms) corners ($\text{PM}^* = 20^\circ$). Diverging runs receive $10^4(1 + \text{lost fraction})$.

4.2. Margin constraint

About hover, DOF i is $\ddot{y} + c_{1,i}\dot{y} + c_{0,i}y = H(s)a$ with $c_1 = D_{l,ii}/M_{ii}$ and the restoring stiffness $c_0 = z_b B/M_{ii}$ in roll and pitch. The actuator path $H(s) = e^{-(n_d+1/2)\Delta ts}/(\tau_m s + 1)$ contains n_d periods of latency, the zero-order hold and the thruster lag. Broken at a , the loop is

$$L_i(s) = G_2(s)(G_1(s) + s)P_i(s), \quad P_i(s) = \frac{H(s)}{s^2 + c_{1,i}s + c_{0,i}}, \quad (6)$$

($L_i = G_1 P_i$ for single-loop structures), with G_1, G_2 the stage transfer functions as realized (Oustaloup). PM_i is the smallest distance of $\arg L_i$ from -180° over all gain crossovers below 300 rad/s; a loop whose crossover lies above (the Nyquist frequency is 314 rad/s) gets $\text{PM}_i = -180^\circ$. The linear analysis predicts the latency at which the simulated loop loses stability (unit test `test_margins_predict_simulated_delay_limit`).

4.3. Search space

Gains are searched as $\log_{10} K \in [-2, \log_{10} 50]$, the orders as $\lambda \in [0, 1.5]$, $\mu \in [0, 1]$, and $n \in [1, 10]$. On a linear gain scale most random initial points violate the margin constraint and all tuners stalled near $J \approx 100$.

5. Tuners and comparison protocol

5.1. Stellar oscillation optimizer

SOO [17] moves a population of N oscillators in two stages per iteration $t = 1, \dots, T$. With the angular frequency $\omega(t) = 2\pi/(P_0 + \Delta P t)$, the scaling factor $S(t) = 2(1 - t/T)$, the best position $\mathbf{x}_b$ and $r_1, r_2, r_3 \sim U(0, 1)$ drawn per component, the first stage forms

$$\mathbf{x}_{o1} = \mathbf{x}_b - r_1 r_3 S(\omega r_1 - 1) (\mathbf{x}_i - |r_1 \sin(r_2) |r_3 \mathbf{x}_b|), \quad (7)$$

$$\mathbf{x}_{o2} = \mathbf{x}_b - r_2 r_3 S(\omega r_1 - 1) (\mathbf{x}_i - |r_1 \cos(r_2) |r_3 \mathbf{x}_b|), \quad (8)$$

$$\mathbf{x}_{new} = r_3 (\mathbf{x}_{o1} + \mathbf{x}_{o2})/2, \quad (9)$$

and the second stage moves each oscillator about the mean $\bar{\mathbf{x}}_3$ of the three best,

$$\mathbf{x}_{osc} = \bar{\mathbf{x}}_3 + \frac{1}{2} [\sin(r\pi)(\mathbf{x}_a - \mathbf{x}_{b'}) + \cos((1-r)\pi)(\mathbf{x}_a - \mathbf{x}_c)], \quad (10)$$

with three other oscillators a, b', c , keeping each component of the old position with probability $1/2$. We follow the equations of the paper as read by the reimplementations in [21]: both stages in every iteration with greedy survivor selection, $P_0 = 3$, $\Delta P = 0.001$, clipping at the bounds. That source documents several differences between the paper and the authors' released code. SOO therefore spends $2N$ evaluations per iteration.

Eqs. (7)–(9) refer to the coordinate origin twice: through $|r_3 \mathbf{x}_b|$ and through the factor $r_3 \in [0, 1]$, which contracts every candidate towards $\mathbf{0}$. As a control we use *BC-SOO*, the same first stage written in coordinates

relative to $\mathbf{x}_b$ (the substitution $\mathbf{x} \mapsto \mathbf{x} - \mathbf{x}_b$),

$$\begin{aligned}\mathbf{x}_{ok} &= \mathbf{x}_b - r_k r_3 S(\omega r_1 - 1)(\mathbf{x}_i - \mathbf{x}_b), \quad k = 1, 2, \\ \mathbf{x}_{new} &= \mathbf{x}_b + r_3((\mathbf{x}_{o1} + \mathbf{x}_{o2})/2 - \mathbf{x}_b),\end{aligned}\tag{11}$$

which is translation invariant by construction.

5.2. Other tuners

PSO with the constriction coefficients of [25] ($w = 0.7298$, $c_1 = c_2 = 1.49618$, velocity limited to 20% of the range), DE/rand/1/bin [12] ($F = 0.5$, $CR = 0.9$), GWO [26], GJO [27] and uniform random search (RS). PSO and DE are the tuners of [10]; GWO and GJO are among the comparison methods of [16].

5.3. Protocol

- *Equal budget.* Every run stops after 3030 evaluations (population 30): 100 iterations for the one-stage methods and 50 for SOO. The equal-iteration protocol of [16], which gives SOO 6030 evaluations, is run as a separate setting (SOO-iter).
- *Mirrored box.* Every tuner is also run on the problem expressed in the coordinates $\mathbf{x}' = \mathbf{lb} + \mathbf{ub} - \mathbf{x}$. The problem is the same; only the position of the coordinate origin relative to the good region changes. A tuner without positional bias gives the same distribution of results in both boxes.
- *Shifted benchmarks.* Sphere, Rastrigin, Rosenbrock and Ackley functions in $D = 22$ on $[-100, 100]^{22}$, with the optimum at the origin and at a random point of $[-60, 60]^{22}$ (15 runs).

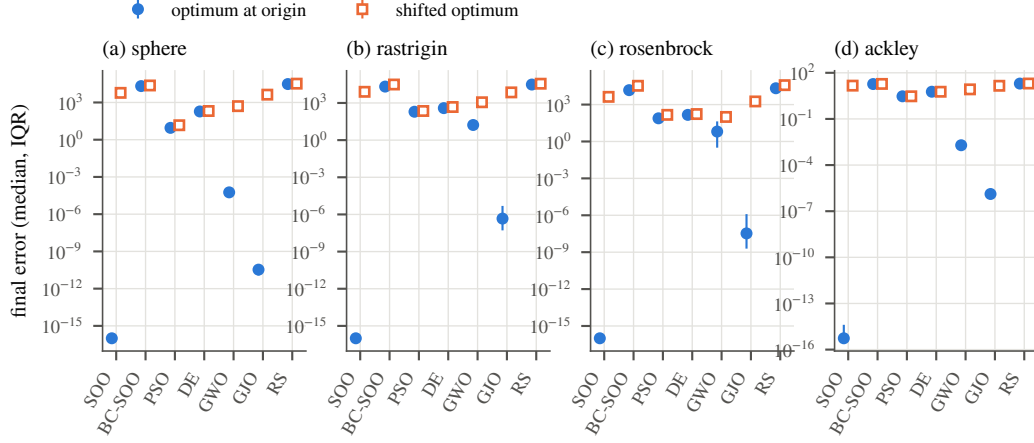


Figure 1: Final error after 3030 evaluations ($D = 22$, box $[-100, 100]^{22}$, 15 runs; median and interquartile range) with the optimum at the origin (filled) and at a random point in $[-60, 60]^{22}$ (open). Values below 10^{-15} are drawn at 10^{-16} .

- Ten independent runs per setting; medians, interquartile ranges and two-sided Mann–Whitney tests.

6. Results

6.1. Positional bias on shifted benchmarks

With the optimum at the origin, SOO reaches 10^{-49} on the sphere and exactly zero on Rastrigin and Rosenbrock; with the optimum shifted it ends at 6×10^3 , 8×10^3 and 4×10^3 (Fig. 1). GJO and GWO show the same pattern with smaller gaps, while PSO and DE give the same errors in both cases, as expected from their translation-invariant updates. BC-SOO is translation invariant but performs like random search: the search ability of SOO comes from its attraction to the origin.

Table 3: Training cost J of the proposed controller for ten runs per tuner (3030 evaluations unless noted). p : two-sided Mann–Whitney tests against PSO (original box) and between the two boxes.

tuner	best	median [IQR], original box	median [IQR], mirrored box	p vs PSO	p boxes
PSO	35.8	40.0 [37.9, 56.2]	40.4 [38.9, 46.1]	–	1.000
DE	54.7	67.0 [58.3, 73.3]	84.4 [56.4, 99.8]	0.003	0.273
GWO	48.5	68.4 [55.9, 113.9]	103.8 [84.9, 119.1]	0.007	0.186
SOO	41.5	85.0 [69.6, 126.4]	59.7 [52.6, 105.3]	0.002	0.385
SOO, 6030 eval.	32.1	60.7 [43.4, 80.3]	–	0.089	–
GJO	57.5	122.3 [90.7, 142.0]	208.0 [165.8, 228.6]	< 0.001	0.002
BC-SOO	215.8	329.8 [275.1, 410.9]	325.3 [186.1, 435.1]	< 0.001	0.521
RS	193.7	403.8 [279.5, 496.1]	363.1 [321.8, 412.6]	< 0.001	0.623

6.2. Tuners on the six-DOF problem

PSO gives the lowest median cost and the same distribution in both boxes (Table 3, Fig. 2). SOO is significantly worse than PSO at the same budget; with the equal-iteration protocol, which doubles its evaluations, the difference is no longer significant ($p = 0.089$) but its median stays 50% higher. GJO depends significantly on the box ($p = 0.002$): its results change when only the position of the coordinate origin relative to the good region changes. On this problem the origin of the search box is not near the optimum (the gains are searched as $\log_{10} K$), so the attraction that makes SOO excel on centred benchmarks no longer helps. PSO is used for the comparison of structures below; DE is kept as the second tuner because it was the best tuner in [10].

6.3. Nominal tuning trades robustness for nominal cost

With the nominal training set and 3030 evaluations, the integer P-PID cascade reaches the lowest best training cost (33.4 against 35.8 for the proposed cascade). With 10 030 evaluations the fractional cascades pass it (best

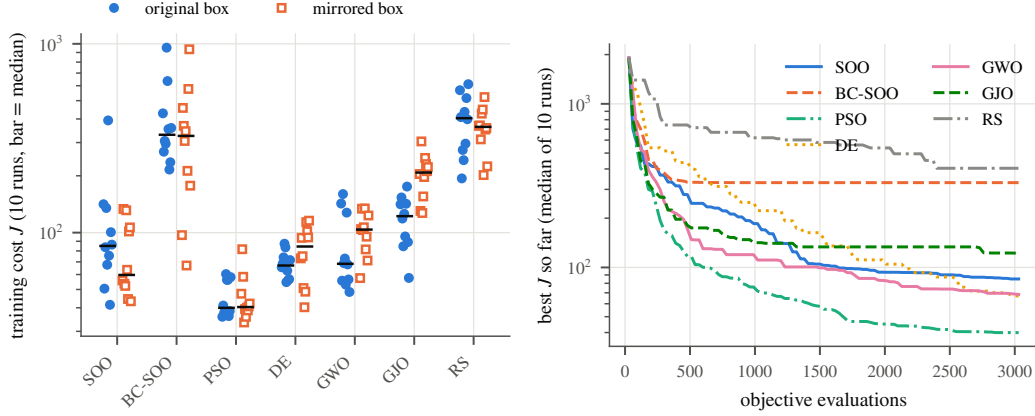


Figure 2: Left: training cost of every run in the original and mirrored boxes (bar: median). Right: median best-so-far cost against the number of evaluations (original box).

29.7 for PI-(1+FOPID), 31.5 for the proposed; median 36.9 for the proposed against 65.0 for P-PID), so the gap at the smaller budget was a search limitation of the 22-parameter structure. The better-trained controllers, however, are more fragile (Fig. 3a): over the 30 runs of each cascade, the rank correlation between training cost and Monte Carlo divergences is -0.92 for P-PID, -0.63 for the proposed cascade, -0.45 for PI-(1+FOPID) and -0.44 for FOPI-FOPD, and only 1–8 of 10 runs per setting are free of divergence (Table 4). The best nominal tuning of the proposed cascade diverges in 14 of 100 draws. The optimizer exploits the nominal cases; robustness has to be part of the cost.

6.4. Robust tuning: controller structures

With the robust training set, none of the 40 cascade tunings diverges in the Monte Carlo study (Fig. 3b). The proposed cascade has the lowest median training cost (80.4), significantly below P-PID (125.3, $p = 0.026$)

Table 4: Training cost J (PSO, ten runs per setting) and number of runs whose controller does not diverge in any of the 100 Monte Carlo draws of T4 (no div.). Nominal: two training cases; robust: four. J is comparable within a column group only. p : Mann–Whitney test of the robust training cost against the proposed cascade.

structure	nominal, 3030		nominal, 10 030		robust, 10 030			p
	median J	no div.	median J	no div.	best J	median J	no div.	
FOPID-(1+TFOID)	40.0	5/10	36.9	4/10	70.6	80.4	10/10	–
PI-(1+FOPID)	52.9	4/10	38.4	1/10	70.0	88.0	10/10	0.473
P-PID	70.1	5/10	65.0	5/10	89.0	125.3	10/10	0.026
FOPI-FOPD	56.2	8/10	50.6	6/10	79.9	127.3	10/10	0.045
FOPID-FOPI	315.6	–	307.2	–	650.9	665.6	–	< 0.001
FOPID	388.9	–	375.0	–	853.2	914.1	–	< 0.001
PID	639.6	–	639.6	–	975.0	975.0	–	< 0.001

and FOPI-FOPD (127.3, $p = 0.045$) and on a par with PI-(1+FOPID) (88.0, $p = 0.47$). In the test cases (Table 5, Fig. 4) its best tuning has the lowest error in the multi-set-point storm test T4 (132.8, 35% below P-PID and 17% below PI-(1+FOPID)), the lowest median and 90th percentile in the Monte Carlo study, and three times less error than P-PID under severe noise (T2). Over all ten runs the differences to P-PID hold: median T4 cost 171.5 against 345.2 ($p = 0.021$) and median T2 cost 14.2 against 33.7 ($p = 0.003$). PI-(1+FOPID) is slightly better in the storm test T3 (50.2 against 52.5) and FOPI-FOPD in T1 and T2, but both are worse in T4 and in the Monte Carlo study. The energy in T4 is the same for all cascades within 2%. The single-loop FOPID of [10], PID and FOPID-FOPI do not diverge either, but they cannot reach a useful bandwidth within the margin constraints: their median training costs are 8–12 times higher (Table 4), their T4 errors 7.8–19 times higher than that of the proposed cascade, and the vehicle oscillates in

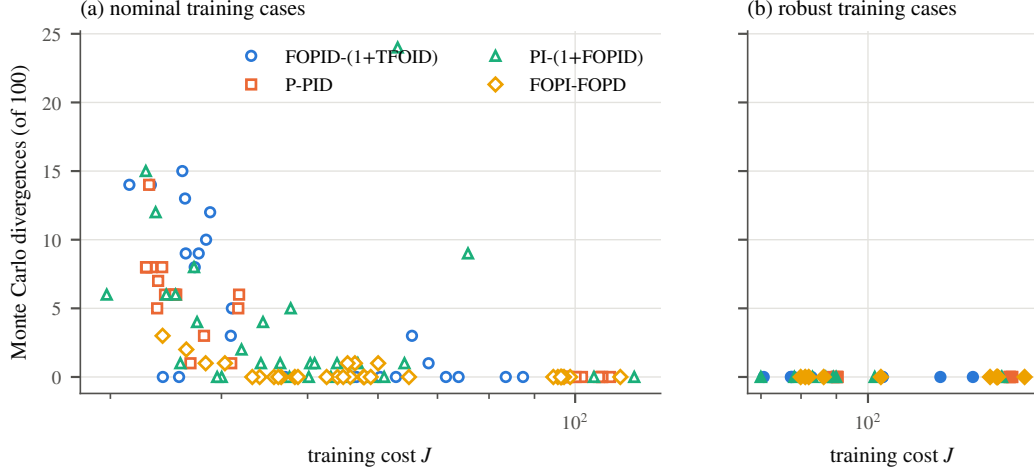


Figure 3: Training cost against the number of Monte Carlo divergences (of 100) for every run of the four cascades. (a) Nominal training set, 3030 and 10 030 evaluations (DE and PSO); (b) robust training set, 10 030 evaluations (PSO). The costs of (a) and (b) are not comparable.

waves (Fig. 5).

6.5. Robustness

With robust tuning, mass, added mass, damping and buoyancy changes move the cost of the cascades by at most 5% (Fig. 6); a slow thruster with doubled latency costs 3–11%. A 12 V battery, which lowers the maximum thrust by 28–29% on the T200 curves, costs 19–37%. Losing a horizontal thruster without telling the allocation still makes the cascades diverge or degrade by more than 20 times, and losing a vertical one costs 2.5–5.3 times. When the failure is known and the allocation is recomputed on the remaining seven thrusters, the cost of the proposed cascade rises by only 1–3% (P-PID 5–11%): the over-actuation of the heavy configuration is useful only together with fault-aware allocation.

Table 5: Test cases for the best robust tuning of each structure. Weighted ITAE as in Eq. (5); E : electrical energy in T4 (60 s); Monte Carlo: T4 with random mass ($\pm 20\%$), added mass and damping ($\pm 50\%$), buoyancy (-2 to $+3\%$), battery 12–20 V, thruster lag 0.05–0.2 s and latency 10–60 ms.

structure (tuner)	weighted ITAE				E T4	Monte Carlo, T4 (100)		
	T1	T2	T3	T4	[kJ]	median	p90	diverged
FOPID-(1+TFOID) (PSO)	3.2	17.0	52.5	132.8	10.6	148	207	0
PI-(1+FOPID) (PSO)	4.4	22.0	50.2	160.1	10.6	172	213	0
FOPI-FOPD (PSO)	2.4	15.2	66.4	163.9	10.8	183	245	0
P-PID (PSO)	4.2	51.7	63.4	203.0	10.8	217	252	0
FOPID-FOPI (PSO)	25.1	37.9	384.6	1031.6	9.0	1093	1652	0
FOPID (PSO)	18.7	224.5	411.8	1039.6	10.0	1108	1662	0
PID (PSO)	16.8	127.4	1119.9	2553.8	10.0	2726	3710	0

6.6. Margins and latency

At the nominal thruster the six loops of the proposed controller (best robust tuning) have phase margins of $45.0\text{--}52.3^\circ$ and tolerate 43–60 ms of additional latency; at the slow corner the smallest margin is 20.2° , on the constraint. To show what the margin constraint buys, the proposed controller was tuned as in the load-frequency study: no latency in the model and no margin requirement (X5, six runs). The training cost drops to 18.6–22.2, but the smallest margin per run is $3.8\text{--}24.7^\circ$. In T4 these controllers reach 75–99 without latency, 180–466 at the nominal 30 ms and 422–1314 at 60 ms, while the robust, margin-constrained controller goes from 133.5 to 138.1 over the same range.

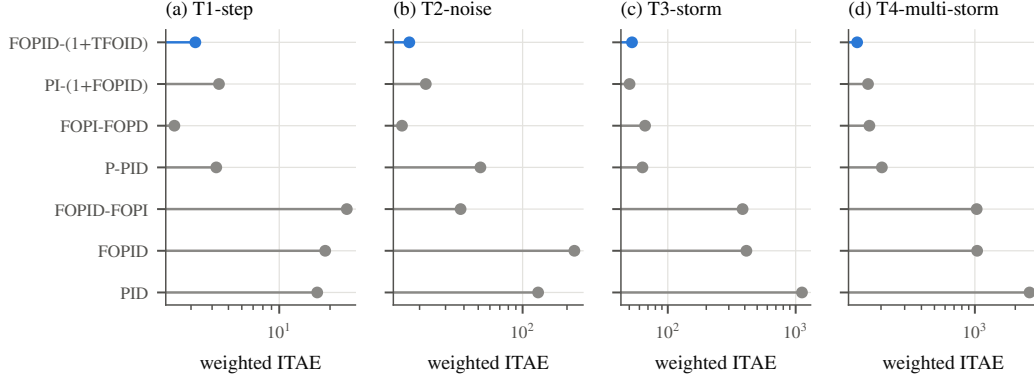


Figure 4: Weighted ITAE in the four test cases, best robust tuning of each structure (logarithmic axes).

7. Discussion and limitations

- *Robustness is a property of the tuning, not only of the structure.* With nominal training cases, every cascade can be tuned to a low nominal cost and a fragile controller; the comparison of structures is meaningful only when the cost contains the robustness requirement. Studies that tune on one nominal scenario (including [10] and [16]) may overstate the advantage of whichever structure their tuner happened to over-fit least.
- *Tuner choice.* The tuners were compared on the nominal cost (X1); PSO was then used for all structures. [TODO: Repeat X1 with the robust training set in the final run.]
- *Thruster dynamics.* No validated dynamic model of the T200 was available; the design covers a range of lag and latency instead. [TODO: Replace the nominal values by those of the MATLAB re-identification]

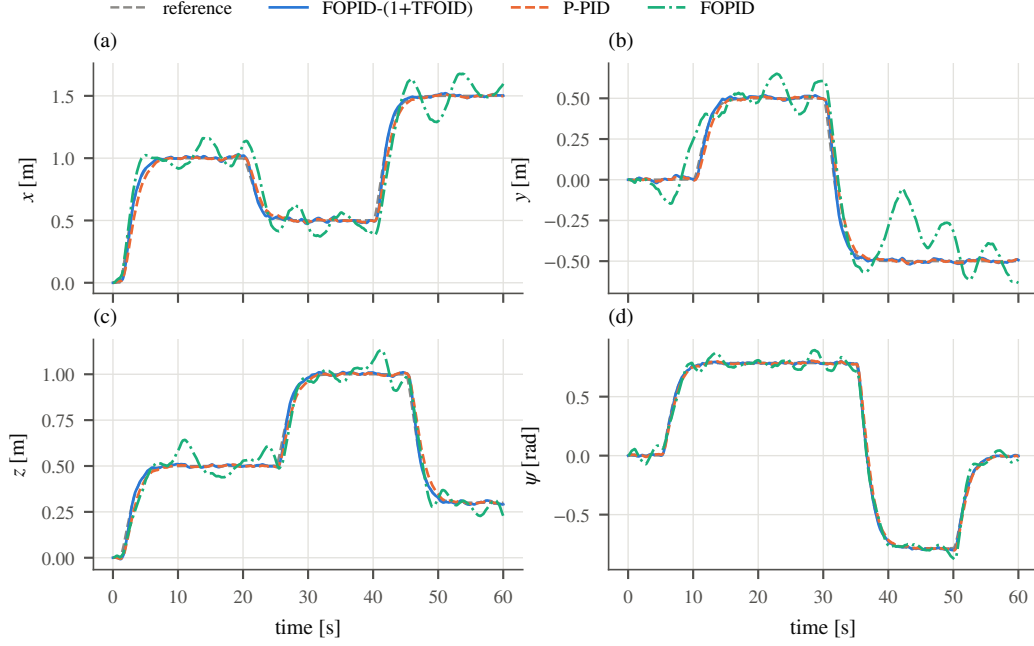


Figure 5: Test T4 (multi-level set points in waves and current): x , y , z and ψ for the proposed cascade, the integer P-PID cascade and the single-loop FOPID of [10], each with its best robust tuning.

and narrow the range accordingly.]

- *Model.* Simulation only; hydrodynamic coefficients from [4]; one parameter set per group of DOFs; margins from a linearization about hover; navigation errors modelled, not estimated. Pool tests or hardware-in-the-loop, as in the FPGA realization of [10], are the next step.

8. Conclusions

We recast the FOPID-(1+TFOID) cascade as a position-velocity controller for all six DOF of an over-actuated BlueROV2 Heavy with data-fitted

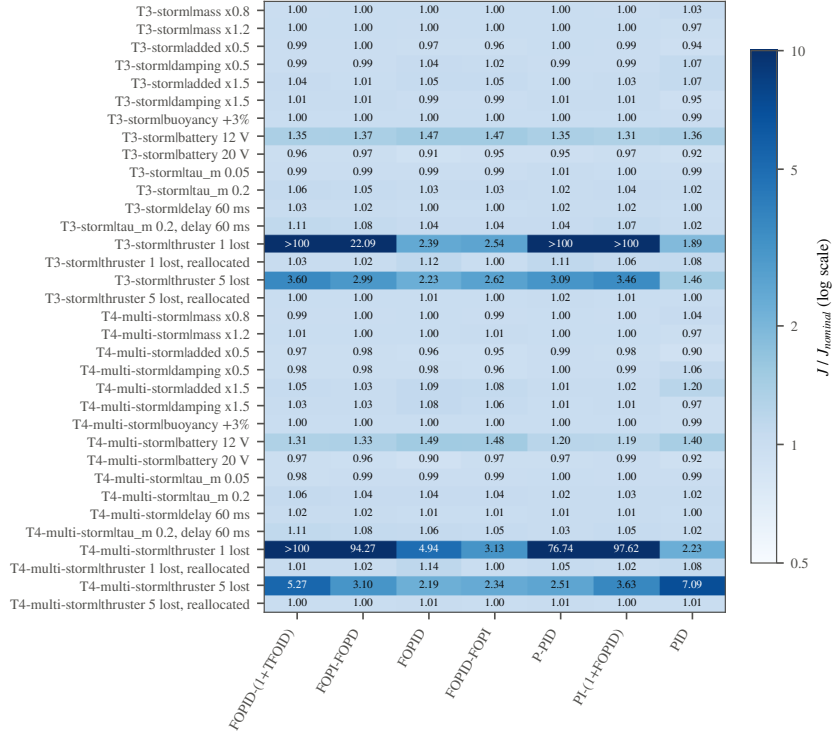


Figure 6: Cost of each variant relative to the nominal test case, per structure, best robust tuning (the controllers are not retuned).

T200 thrusters and tuned it under phase-margin constraints over a range of thruster lag and latency. Compared at an equal budget and in a mirrored search box, PSO was the best tuner, while the recent SOO and GJO showed a strong attraction to the coordinate origin. Tuning on nominal cases produced controllers whose nominal cost and Monte Carlo robustness were inversely related; with a robust training set no tuning of any cascade diverged, and the proposed cascade became the best structure, with 35% less error than the integer P-PID cascade in the multi-set-point storm test and three times less under severe sensor noise. A known thruster failure costs it 1–3% when

the allocation is recomputed.

Data and code availability

Code, results and figure scripts: [\[TODO: repository URL\]](#). The T200 data are fetched from the manufacturer and checked by SHA-256 [23].

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